

PHYS 5BL–Spring 2026 — Prelab 7 Response

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February 18, 2026

1. We have $\ddot{x} + \beta\dot{x} + \omega^2x = 0$. Overdamping means our solutions to the equation are real negative exponentials; that is, $\beta^2 > 4\omega^2$ and our solution is $c_1e^{r_1} + c_2e^{r_2}$ where $r_1, r_2 = \frac{-\beta \pm \sqrt{\beta^2 - 4\omega^2}}{2}$. As these functions are assuredly exponentials of negative reals, the function is necessarily one-to-one and can only pass through any given point (including the equilibrium point) once.
2. (a) In an underdamped oscillator, $\beta^2 < 4\omega^2$ and our general solution follows $Ae^{-\alpha t} \cos(\omega t) + Be^{-\alpha t} \sin(\omega t)$. This is equivalent to $Ce^{-\alpha t} \cos(\omega t + \phi)$ for initial amplitude $C = \sqrt{A^2 + B^2}$ and phase shift $\phi = \arctan \frac{B}{A}$. The only component capable of equalling zero is the $\cos(\omega t)$ term, so the zero crossings will be wholly determined by the sinusoidal part of the solution. Alternatively phrased, $Ae^{-\alpha t}$ is always real so only the $\cos(\omega t)$ is capable of determining the argument of the solution, and thus the solution will cross zeroes when the sinusoidal part of the solution does. However, a zero crossing occurs every half-period.
(b) Yes, they represent the "far" and "close" parts of the oscillatory path as it decays, and occur once every period.